

Emission Aware Multi-Metric Reinforcement Learning for Adaptive Traffic Signal Control in Sri Lanka

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Abstract—Urban intersections in Sri Lanka experience growing congestion and greenhouse gas emissions, while signal control remains fixed-time and insensitive to demand variability. An adaptive traffic signal control framework based on reinforcement learning (RL) is proposed. The control task is formulated as a Markov decision process with state features comprising queue length, waiting time, braking events, and estimated emissions. A Deep Q-Network (DQN) is trained in the SUMO–TraCI microsimulation under static and dynamic demand scenarios. Multiple reward formulations are examined, including emission-only and multi-metric designs with normalized, weighted components. Experiments show that emission-only rewards yield unstable learning, increased queuing, and no measurable emission reduction, indicating the limitations of single-metric optimization. A balanced reward combining queue length and braking achieves the most favorable trade-off, reducing average delay by up to 22% and CO₂ emissions by 15% relative to fixed-time baselines. The framework is further instantiated on OpenStreetMap-derived models of Sri Lankan intersections (Borella, Kollupitiya), demonstrating applicability to local geometries and operating conditions. The results indicate that emission-aware RL controllers can outperform fixed-time operation and offer a viable pathway for deployment to support congestion mitigation and emission-reduction objectives.

Index Terms—Traffic, DQN, Emission, Reinforcement learning, SUMO, Signal

I. INTRODUCTION

Urban traffic congestion remains a major challenge, impacting mobility, fuel consumption, and environmental sustainability. Growing vehicle ownership and limited road capacity exacerbate delays, while road transport continues to emit significant amounts of CO₂, NO_x, and hydrocarbons [1]–[3]. Since traffic signals directly influence both mobility and emissions, their effective control is crucial [4][5].

Conventional systems—fixed-time, actuated, and adaptive—offer varying trade-offs. Fixed-time control is economical but insensitive to changing demand [4][6], while actuated systems use detectors for improved responsiveness, though mainly at isolated intersections [7].

Adaptive frameworks like SCOOT (UK) and SCATS (Australia) adjust timings in real time, reducing delays but requiring heavy infrastructure and focusing primarily on

mobility rather than emissions [5][8][9]. Modern systems such as SURTRAC and InSync show improved efficiency but remain mobility-centric [10][11].

Reinforcement Learning (RL) offers a promising alternative, with Deep Q-Networks (DQNs) capable of learning optimal signal policies directly from traffic interaction without predefined rules [5][12]. However, most RL studies optimize only mobility metrics such as delay or throughput [13][14], overlooking emissions caused by stop-and-go driving and cold starts [1]–[3][15][16].

This study presents an emission-aware adaptive traffic signal control framework that integrates a custom reward function combining queue length, braking, idling, and emissions. Developed and tested in the Simulation of Urban Mobility (SUMO) platform under static and dynamic traffic scenarios, it is further evaluated for deployment in Colombo, Sri Lanka. By coupling mobility and environmental objectives, the framework demonstrates the potential of RL-based control to reduce congestion and emissions, supporting sustainable urban traffic management.

II. LITERATURE REVIEW

A. Conventional Traffic Signal Control

Conventional traffic signal control methods are categorized into fixed time, actuated, and adaptive approaches. Fixed-time systems operate on predetermined cycles derived from historical surveys, such as Webster’s formula [4]. Widely deployed in cities such as New York and London, they often employ coordinated “green waves” to enhance progression [17]. While cost-effective and predictable, their inability to adapt to real-time fluctuations reduces efficiency under varying demand [5].

Actuated control improves responsiveness by employing detectors (e.g., loop sensors, cameras) to extend or terminate phases dynamically. Semi-actuated systems prioritize main roads, while fully actuated systems adjust all phases in real time. Common in the United States and Europe [13], these systems reduce unnecessary delay but are largely optimized locally and require significant investment for detectors and maintenance [14][5].

Adaptive systems adjust timing across networks using real-time data. SCOOT in the United Kingdom [18] and SCATS in Australia [8][9] have achieved 10–30% reductions in delay and travel time. More recent frameworks, such as InSync [11] and SURTRAC in the United States [10], incorporate artificial intelligence and decentralized optimization to enhance efficiency. However, adaptive control requires substantial infrastructure and reliable communication systems, restricting widespread deployment [12]. In summary, while conventional methods have advanced from static to adaptive systems, challenges of scalability, cost, and responsiveness persist, motivating the exploration of reinforcement learning (RL)–based approaches.

B. Traffic Signal Control in Sri Lanka

In Sri Lanka, most traffic signals operate on fixed-time control, with green, amber, and red phases preset and stored in programmable memory chips. Widely deployed across Colombo under University of Moratuwa initiatives, this low-cost approach lacks adaptability to fluctuating demand. The implementation process typically involves junction selection, surveys, timing design, programming, and operation, as illustrated in Fig. 1.

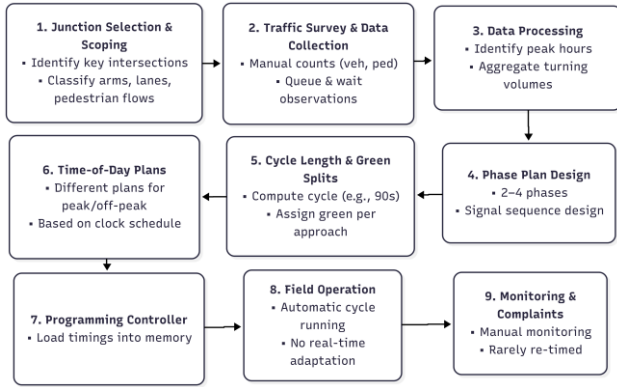


Fig. 1. Sri Lanka's fixed-time traffic signal control process

Although simple and inexpensive, fixed-time systems require frequent recalibration, and modifications at one junction often disrupt coordination with adjacent intersections. They also fail to respond to congestion, pedestrian demand, or emergencies, causing inefficiency and higher emissions.

Recent work at NSBM Green University proposed an adaptive framework integrating YOLOv8 for vehicle detection and Long Short-Term Memory (LSTM) networks for traffic prediction, showcasing the potential of machine learning–based control [19]. However, most intersections in Colombo remain static, emphasizing the need for adaptive, emission-aware traffic management solutions.

III. EMISSIONS AND TRAFFIC FLOW DYNAMICS

Traffic conditions strongly influence vehicular emissions. Congestion, stop-and-go patterns, and ineffective signal control increase fuel use and pollutant output, with stop-and-go driving and cold-start operation representing two critical mechanisms.

A. Conventional Traffic Signal Control

Repetitive braking and acceleration cause momentum loss and fuel-intensive recovery, producing spikes in CO₂ emissions [13]. Engines are most efficient at steady speeds

(40–60 km/h); irregular cycles reduce combustion efficiency [13]. Idling during queues further elevates emissions [12], while frequent halts cool exhaust gases, reducing catalytic converter activity [1].

B. Cold-Start Emissions and Catalyst Warm-Up

Catalytic converters are largely inactive below ~250–300 °C, achieving effective pollutant conversion only after the ~400 °C light-off threshold [2][4]. Studies show that 60–90% of CO and hydrocarbon emissions occur during the initial minutes of operation [3][15]. The problem is intensified in short trips and congested traffic, where idling prevents sufficient heating. Even after light-off, prolonged idling may cool catalysts, limiting conversion efficiency [16][20].

C. Implications for Traffic Signal Control

Fixed-time signal plans, which induce unnecessary stops and extended idling, exacerbate both momentum losses and catalyst inefficiencies [12][16]. Adaptive control can mitigate these effects by smoothing flow and minimizing idle periods, keeping vehicles within efficient operating regimes. Reinforcement learning–based frameworks should therefore employ reward functions that penalize braking, waiting, and CO₂ emissions to optimize traffic efficiency and sustainability.

IV. METHODOLOGY

This research proposes a Reinforcement Learning (RL)–based adaptive traffic signal control system to optimize traffic flow and reduce greenhouse gas (GHG) emissions at Malabe Junction, Sri Lanka. The methodology comprises data integration, simulation modeling, RL training, and evaluation.

A. Simulation Environment

The framework was evaluated in SUMO, an open-source microscopic traffic simulator. A four-way intersection with one lane per direction was modeled using SUMO XML files for nodes, edges, and signal logic to ensure realistic vehicle dynamics and control.

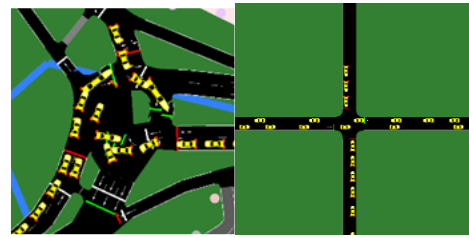


Fig. 2. Four-way intersection network

B. Traffic Scenarios

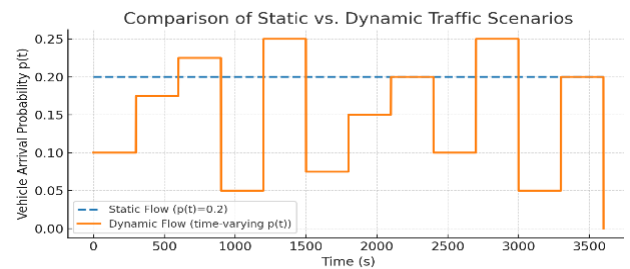


Fig. 3. Static vs. dynamic traffic scenarios

Two traffic demand scenarios were developed to assess the performance of the reinforcement learning agents. In the static

scenario, vehicle arrivals followed a constant probability of $p(t)=0.2, \forall t \in [0,3600]$ for all four approaches (north–south, south–north, east – west, and west – east), representing steady-state traffic with uniform demand. In the dynamic scenario, arrival probabilities varied within $p(t) \in [0.025,0.25]$ updated every 300 s to capture temporal fluctuations typical of rush hours, congestion buildup, and low-demand periods. Thus, the static scenario served as a simplified baseline, while the dynamic scenario approximated realistic and time-varying traffic conditions.

C. Reinforcement Learning Problem Formulation

The traffic signal control problem is framed as a Markov Decision Process (MDP), enabling the application of reinforcement learning. At each timestep t , the agent observes the state of the intersection, selects a signal phase, and receives a reward reflecting both traffic efficiency and environmental performance. The MDP is defined as

$$M = \langle S, A, P, R, \gamma \rangle \quad (1)$$

where S is the state space, A the action set, P the transition dynamics governed by SUMO, R the reward function, and $\gamma \in (0,1]$ gamma the discount factor.

The state space captures traffic and environmental conditions through features such as queue length, waiting time, braking events, and emissions. The action space corresponds to feasible traffic signal phases, with operational constraints (e.g., yellow and minimum green times) enforced by the simulator. At each decision step, the agent receives a scalar reward reflecting control performance; the detailed reward design, including normalization and weighting of metrics, is described in Section 5.

The agent's objective is to maximize the expected discounted return

$$J(\pi_\theta) = \{E\}_{\{\pi_\theta\}} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right] \quad (2)$$

where $\pi_\theta(a|s)$ denotes the policy parameterized by θ . This formulation enables the learning of adaptive traffic signal policies that respond to dynamic traffic conditions while incorporating sustainability objectives.

D. Deep Q-Network (DQN) Implementation

To model the adaptive traffic signal control agent, a Deep Q-Network (DQN) framework was employed. The DQN algorithm extends classical Q-learning by leveraging a neural network to approximate the action-value function, enabling effective policy learning in high-dimensional and stochastic environments.

1. Neural network architecture.

The Q-function was approximated using a Multi-Layer Perceptron (MLP) for its efficiency and suitability for discrete action spaces. The model comprised two hidden layers with 64 neurons each and nonlinear activations, and an output layer with four neurons representing the possible signal phases (e.g., north–south green/yellow, west–east green/yellow). This simple yet effective design ensured stable convergence without overfitting.

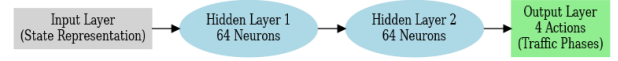


Fig. 4. DQN neural network for traffic signal phase decisions

2. Training parameters and optimization.

The network was trained using the Adam optimizer ($\alpha = 1 \times 10^{-4}$) with a mini-batch size of 64 and a replay buffer of 2000. An ϵ -greedy policy decayed ϵ from 1.0 to 0.1. The target network was updated every 10 000 steps, and training began after 720 steps. A discount factor $\gamma = 0.99$ was used to favor long-term rewards.

E. Custom Reward Design

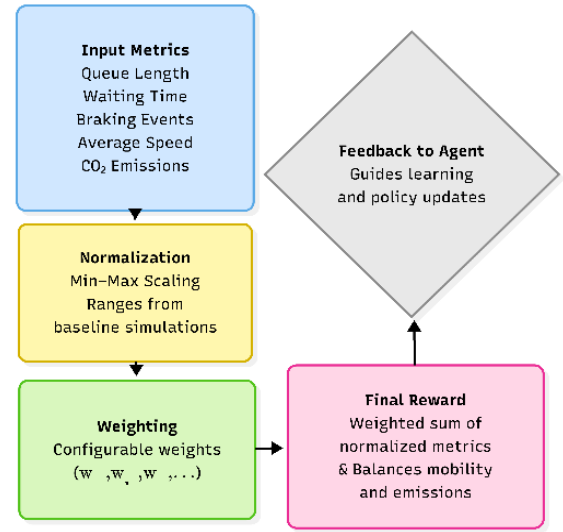


Fig. 5. Workflow of reward design

The reward function defines the optimization objective of the RL agent. Traditional traffic control typically focuses on single mobility metrics (e.g., queue length or delay), neglecting environmental factors. This study employs a custom multi-metric reward combining mobility (queue length, waiting time, braking events, average speed) and environmental (CO₂ emissions) indicators. All metrics were normalized for scale consistency and aggregated through a weighted scheme to produce the final scalar reward, which guided agent learning (see Fig. 5).

Reward design workflow illustrates the process of combining input traffic metrics through normalization and weighting into a final reward signal, which guides the RL agent.

Formally, the reward at timestep t is defined as:

$$R_t = \sum_{i=1}^n w_i m_{i,t} \quad (3)$$

where $m_{i,t}$ is the normalized value of metric i , and w_i is its weighting factor. Normalization ranges were derived from baseline fixed-time simulations to provide consistent scaling, while weight configurations were explored using a line-search strategy to evaluate trade-offs between congestion reduction and emission mitigation.

Two reward variants were considered: (i) a Mobility-Oriented Reward, which emphasizes queue and braking

reduction, and (ii) an Emission-Aware Reward, which extends the mobility formulation by penalizing emissions. These two designs enabled evaluation of whether sustainability objectives can be integrated into reinforcement learning without compromising traffic efficiency.

F. Baseline Controllers

Two baselines were implemented: (i) a fixed-time controller with a predetermined cycle of signal phases, and (ii) a cyclic controller (Cyclic Agent) that rotates phases at fixed intervals. Both operate without state awareness and thus cannot adapt to real-time traffic conditions. Their performance under static and dynamic demand scenarios provides benchmark references for evaluating the improvements achieved by the reinforcement learning agents.

G. Simulation to Deployment in Sri Lankan Intersections



Fig. 6. SUMO simulation of Kollupitiya and Borella junctions

Following validation on a simplified intersection, the framework was adapted for real Sri Lankan networks (Fig. 6). The workflow proceeds from network construction to performance evaluation. OpenStreetMap (OSM) data were converted into SUMO networks to replicate junction geometry and signal phases at sites such as Kollupitiya and the Galle Road corridor. Traffic demand scenarios—static, dynamic, and peak flows—were generated to train the agent under diverse conditions.

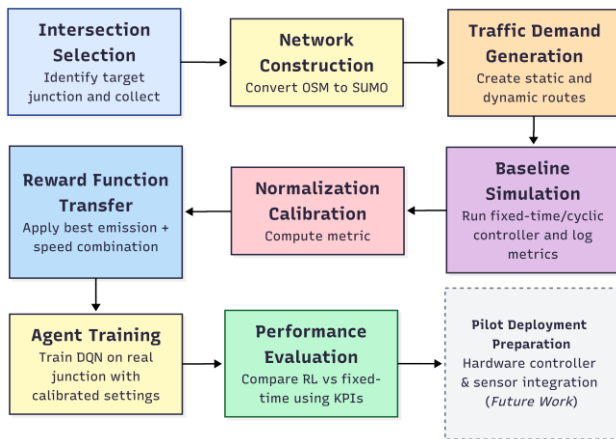


Fig. 7. Deployment of DQN-based traffic signal control

Baseline fixed-time simulations provided benchmarks for normalization of waiting time, braking, queue length, emissions, and speed. The optimized multi-metric reward identified earlier was applied, and the DQN agent was trained and evaluated against fixed-time baselines. Key metrics—

emissions, average speed, and waiting time—were monitored to assess performance.

Although this study concludes with simulation-based validation, future work will focus on real-world deployment involving hardware integration, roadside sensors, and communication protocols.

1) Limitations: The current framework relies on synthetic demand within SUMO rather than field traffic data, which may limit generalization. Future research will incorporate empirical counts for improved calibration and validation.

V. RESULTS

A. Static Traffic Flow

The DQN agent was evaluated against fixed-time controllers with 30 s, 60 s, and 150 s cycle lengths under static traffic conditions. Performance was measured in terms of CO₂ emissions, average speed, stopped vehicles, and aggregated metrics (Figs. 8–10, Table I).

1. CO₂ Emissions

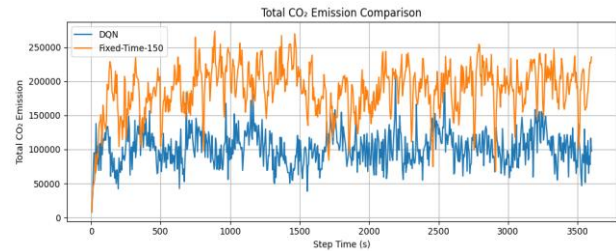


Fig. 8. CO₂ emissions comparison under static flow

The DQN consistently produced lower CO₂ emissions than all fixed-time baselines (Fig. 8). The 150 s cycle yielded the highest values due to prolonged red phases causing idling and abrupt acceleration, while the DQN adaptively allocated green times to reduce unnecessary stops.

2. Average Speed

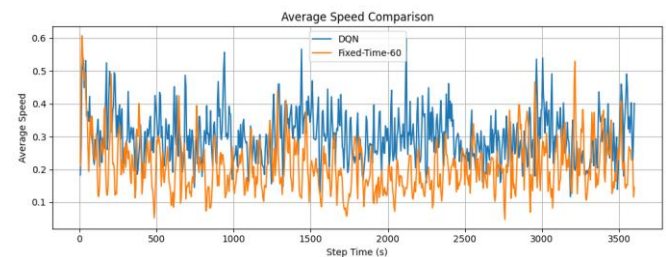


Fig. 9. Average speed comparison (DQN vs. Fixed-60) in static flow

As shown in Fig. 9, the DQN maintained higher and more stable average speeds than the 60 s cycle. Adaptive phase timing enabled smoother flow and reduced halts typical of fixed-time operation.

3. Quantitative Results

TABLE I. AVERAGE PERFORMANCE METRICS FOR DQN FIXED-TIME

Controller	Avg. Reward	Avg. Speed	Avg. CO ₂ Emission	Avg. Wait Time	Avg. Brake	Avg. Stopped Vehicles
DQN	-0.699	0.301	102,225	84.3	-14.31	7.11
Fixed 30	-0.713	0.287	108,399	126.3	-13.62	8.64
Fixed 60	-0.789	0.211	142,055	570.3	-12.65	19.96
Fixed-150	-0.858	0.142	187,763	2495.4	-9.37	40.23

Table I presents the average performance metrics. The DQN achieved the best overall reward (-0.699), highest average speed (0.301), and lowest CO₂ emissions ($102,225$). Waiting time decreased to 84.3 s, compared with 126.3 s (Fixed-30) and 2495.4 s (Fixed-150). Average stoppages were 7.1 vehicles, lower than 8.6 , 20.0 , and 40.2 , respectively. Although braking slightly increased (-14.3 vs. -13.6 for Fixed-30), this minor trade-off was outweighed by significant reductions in delay, congestion, and emissions. Overall, the DQN consistently outperformed all fixed-time controllers, demonstrating superior efficiency and environmental performance under static flow conditions.

B. Emission-Only Reward

To examine the effect of reward design, the DQN agent was trained with CO₂ emissions as the sole reward metric. Results are shown in Figs. 11–12 and Table II.

1) Average Speed

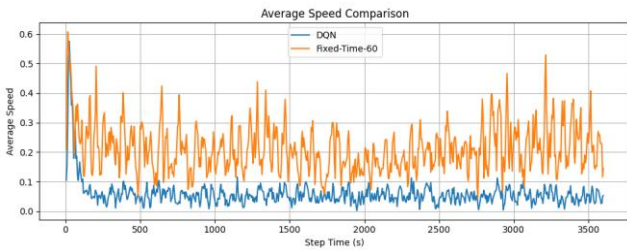


Fig. 10. Avg speed comparison in emission-only reward scenario

Fig. 11 shows that the DQN speed collapsed to 0.059 , far below all fixed-time baselines. The agent effectively stalled traffic instead of enabling flow.

2) CO₂ Emissions

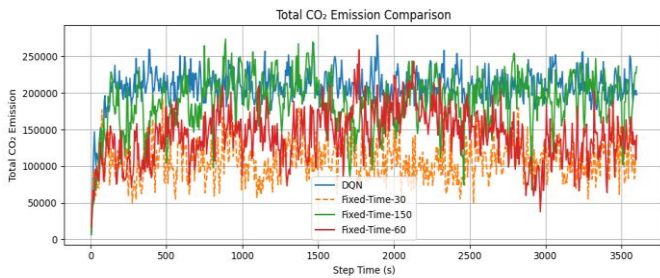


Fig. 11. CO₂ emissions in emission-only reward scenario

Unexpectedly, DQN produced the highest emissions ($207,843$), exceeding even the worst-performing 150 s baseline ($187,763$) (Fig. 12). This demonstrates that emissions cannot be minimized in isolation, as they depend on throughput, idling, and acceleration dynamics.

3) Quantitative Results

TABLE II. AVERAGE PERFORMANCE METRICS FOR DQN FIXED-TIME

Controller	Avg. Reward	Avg. Speed	Avg CO ₂ Emission	Avg. Wait Time	Avg. Brake	Avg. Stopped Vehicles
DQN	-	0.059	207,843	113,317	-1.95	64.87
Fixed 30	-0.713	0.287	108,399	126.3	-13.62	8.64
Fixed 60	-0.789	0.211	142,055	570.3	-12.65	19.96
Fixed 150	-0.858	0.142	187,763	2495.4	-9.37	40.23

Table II confirms the failure of the emission-only reward across all metrics. Waiting time reached $113,317$ s, several orders of magnitude higher than any baseline. Although braking activity appeared lower (-1.95 vs. -13.62 for Fixed-30), this was an artifact of immobility rather than smoother flow.

The emission-only reward design produced unstable learning, gridlock, and increased emissions. These results highlight the inadequacy of single-objective reward functions and confirm the necessity of multi-metric formulations for stable and effective RL-based traffic control.

VI. DISCUSSION

The results emphasize the critical role of reward design in reinforcement learning (RL)-based traffic signal control.

A. Multi-Metric Rewards

A balanced reward integrating mobility metrics (queue length, waiting time, braking) with CO₂ emissions enabled the DQN agent to allocate green times adaptively, reduce stoppages, and lower emissions relative to fixed-time control, while maintaining stability across objectives.

B. Emission-Only-Reward

Optimizing solely for CO₂ yielded poor results—unstable learning, near-zero speeds, and heavy congestion. Emissions were not reduced compared to the Fixed-30 baseline, as CO₂ depends on speed, acceleration, and flow conditions, which cannot be optimized independently.

C. Deployment Implications

Effective real-world control requires multi-objective reward shaping that balances emissions, mobility, and safety. Deployment must also enforce constraints such as minimum green times to comply with safety standards.

- (1) Takeaway: Multi-objective rewards are essential for RL-based traffic control, enabling simultaneous gains in efficiency and sustainability and supporting deployment in urban intersections.

VII. CONCLUSION

This study presented a reinforcement learning (RL)-based adaptive traffic signal control framework to mitigate congestion and greenhouse gas (GHG) emissions at Sri

Lankan intersections. The problem was modeled as a Markov Decision Process (MDP), with a Deep Q-Network (DQN) trained in the SUMO–TraCI environment under static and dynamic conditions. Multiple reward formulations were evaluated, including emission-only and multi-metric approaches.

Emission-only rewards proved ineffective, resulting in unstable learning, severe congestion, and no meaningful emission reduction. In contrast, multi-metric rewards, particularly those integrating queue length and braking, achieved stable learning and balanced improvements, reducing average vehicle delay by up to 22% and CO₂ emissions by 15% compared to fixed-time baselines.

The framework was further applied to real intersections (Malabe, Borella, Kollupitiya) using OpenStreetMap-based SUMO models, demonstrating deployment feasibility. Overall, the findings confirm that multi-objective reward design is critical for effective RL-based traffic control and underscore its potential to replace fixed-time systems while advancing sustainable mobility and climate goals.

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